Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
M.Sc. Computer Science-II

Title of the Course: Full Stack Development								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-13	MS-CS631T	04	00	04	60	30	70	100

Essential prerequisites

1. Students should be familiar with concepts such as variables, data types, loops, conditionals, and functions.
2. Students should understand the structure of web pages, how to create responsive designs, and have a basic knowledge of how the web works (HTTP, URLs, etc.).
3. Students should understand basic Git commands, branching, and collaboration using platforms
4. Students should be comfortable using the command line interface, navigating directories, and performing basic tasks using commands like `cd`, `ls`, and `mkdir`.

Learning Objectives:

1. Students should be able to create responsive and interactive user interfaces using modern front-end frameworks such as React, Angular, or Vue.js.
2. Students should be capable of designing and implementing server-side applications using a server-side language (e.g., Node.js, Python, Ruby) and connecting them to databases (SQL or NoSQL).
3. Students should be capable of integrating front-end and back-end components seamlessly, handling data flow, managing state, and ensuring the overall functionality of a complete web application.
4. Students should follow industry-standard coding conventions, utilize version control effectively (Git/GitHub), and implement secure and scalable development practices, including deployment and containerization.

Course Outcomes (Cos):

1. Students will demonstrate proficiency in both front-end and back-end development, including the ability to design and implement responsive user interfaces, build robust server-side applications, and integrate these components into a full-stack application.
2. Students will apply industry-standard development tools and best practices, including version control with Git, deployment strategies, containerization, and continuous integration/continuous deployment (CI/CD), to efficiently develop, test, and deploy web applications.

3. Students will develop strong problem-solving and critical-thinking skills, enabling them to analyze and solve complex challenges in full-stack development. They will be able to apply logical reasoning to troubleshoot issues, optimize performance, and implement secure and scalable solutions.
4. Students will have experience working on collaborative projects, demonstrating effective communication, teamwork, and the ability to contribute to a development team. They will showcase their ability to manage project tasks, use project management tools, and contribute to a shared codebase through version control.

Detailed Syllabus:

Unit	Course Contents	Allotted Hours
UNIT 1	Front-End Development: 1. HTML5 and CSS3: - Semantic HTML - CSS layout techniques - Responsive design and media queries 2. Version Control: - Git and GitHub - Branching and merging - Pull requests	**08**
UNIT 2	Back-End Development: 1. Server-Side Language: - Node.js (JavaScript) or Python (Django/Flask) or Ruby (Rails) - Server setup and configuration 2. Databases: - SQL (e.g., PostgreSQL, MySQL) - NoSQL (e.g., MongoDB) - ORM (Object-Relational Mapping) 3. APIs and RESTful Services: - Designing and building APIs - REST principles - Authentication and Authorization 4. Server-Side Framework: - Express.js (for Node.js) - Flask/Django (for Python) - Ruby on Rails (for Ruby)	**12**
UNIT 3	Full-Stack Development: 1. Web Security: - HTTPS, SSL - Cross-Site Scripting (XSS) and Cross-Site Request Forgery (CSRF) prevention	**12**

- Authentication best practices (JWT, OAuth)

2. Web Performance:
   - Page load optimization
   - Caching strategies
   - Front-end optimization techniques
3. Containerization and Deployment:
   - Docker
   - Container orchestration (e.g., Kubernetes)
   - Deployment on cloud platforms (AWS, Azure, or Google Cloud)

UNIT 4 DevOps and Tools: **08**

1. Continuous Integration/Continuous Deployment (CI/CD):
   - Jenkins, GitLab CI, Travis CI
   - Automated testing
2. Package Managers:
   - npm (for Node.js), pip (for Python), gem (for Ruby)
3. Web Development Tools:
   - Webpack, Babel
   - Browser Developer Tools

UNIT 5 Soft Skills and Collaboration: **08**

1. Agile Development:
   - Scrum or Kanban methodologies
   - User stories, sprints, retrospectives

2. Collaboration Tools:
   - JIRA, Trello, Slack
   - Communication and teamwork

UNIT 6 Additional Electives (Based on Job Requirements): **12**

1. Mobile Development:
   - React Native or Flutter (for cross-platform)
   - Swift (for iOS) or Kotlin (for Android)

2. Data Science/Analysis:
   - Python libraries (NumPy, Pandas)
   - Data visualization (Matplotlib, D3.js)

3. Testing:
   - Unit testing (Jest, Mocha)
   - End-to-end testing (Selenium, Cypress)

Suggested Readings/Material:

1. Mastering Nod.js, Sandro Pasquali, Packt
2. Node.js complete reference guide, Velentin Bojinov, David Herron, Dioge Resende, Packt

3. Smashing Node.js, Java Script Everywhere, Guillermo Rauch, WILEY
4. The Devops Additional Playbook, Sanjiv Sharma.
5. Professional Git, Brenyt Laster.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar

(Autonomous)

Syllabus

M.Sc. Computer Science-II

Title of the Course: Machine Learning								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-14	MS-CS632T	04	00	04	60	30	70	100

Learning Objectives:

- To be able to apply machine learning algorithms to solve problems of moderate complexity.
- To apply the algorithms to a real-world problem, optimize the models learned and report on the expected accuracy that can be achieved by applying the models.

Course Outcomes:

Upon completing this course, each student will be able to:

- Recognize the characteristics of machine learning that make it useful to real-world problems.
- Process available data using python libraries and predict outcomes using Machine Learning algorithms to solve given problem.
- Able to estimate Machine Learning models efficiency using suitable metrics.

Unit	Course Contents	Allotted Hours
Unit I	**Introduction to Machine Learning** 1.1 Data Science, Artificial Intelligence and Machine Learning Why Learn and What is Learning, What is Machine Learning Traditional Programming Vs. Machine Learning, Machine 1.2 Learning Process, Types of Data, Key Elements of Machine Learning (Representation, Evaluation and Optimization), Dimensionality Reduction (Feature Reduction) 1.3 Descriptive and Inferential Statistics: Probability, Distribution, Distance Measures (Euclidean and Manhattan), Correlation and Regression, Hypothesis Testing. Creating our own dataset, Importing the dataset, Handling 1.4 Missing Data, Splitting the dataset into the Training set and Test set, Feature Scaling.	(07)

Unit II	**Machine Learning Models** 2.1 Type of Learning- Supervised, Unsupervised and Semi-Supervised Learning. 2.2 Components of Generalization Error (Bias, Variance, under fitting, overfitting) 2.3 A Learning System Cycle and Design Cycle. 2.4 Metrics for evaluation viz. accuracy, scalability, squared error, precision and recall, likelihood, posterior probability. 2.5 Classification Accuracy and Performance.	(06)
Unit III	**Regression Models** **3.1** Linear Regression - Simple, Multiple, Polynomial **3.2** Non-linear Regression – Decision Tree, Support Vector, Random Forest.	(10)
Unit IV	**Classification Models** **4.1** K – Nearest Neighbors (KNN) **4.2** Logistic Regression **4.3** Naive Bayes Theorem **4.4** Support Vector Machine **4.5** Decision Forest Classification **4.6** Random Tree Classification	(10)
Unit V	**Clustering Models** **5.1** K-means **5.2** Hierarchical Clustering(Agglomerative, Divisive), Dendrogram **5.3** Selecting optimal number of clusters: Within Clusters Sum of Squares (WCSS) by Elbow Method	(08)
Unit VI	**Association Rules** **6.1** Key Terms: Support, Confidence and Lift **6.2** Apriori Algorithm	(04)
Unit VII	**Reinforcement Learning** **7.1** Upper Confidence Bound **7.2** Thompson Sampling **7.3** Q-Learning	(07)
Unit VIII	**Deep Learning** **8.1** Artificial Neural Network **8.2** Convolution Neural Network **8.3** Recurring Neural Network **8.4** Convolution Neural Network **8.5** Recurring Neural Network	(08)

Suggested Readings

1. Mitchell, Tom M. "Machine learning. WCB." (1997).
2. Rogers, Simon, and Mark Girolami. A first course in machine learning. CRC Press, 2015.
3. Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. The elements of statistical learning. Vol.1.Springer, Berlin: Springer series in statistics, 2001.
4. Witten, Ian H., and Eibe Frank. Data Mining: Practical machine learning tools and techniques. MorganKaufmann, 2005.
5. Machine learning course material by Andrew Ng, Stanford university

6. Sutton, Richard S., and Andrew G. Barto. Reinforcement learning: An introduction. Vol. 1. No. 1. Cambridge: MIT press, 1998.
7. Iba, Takashi, et al. "Learning patterns: A pattern language for active learners." Conference on PatternLanguages of Programs (PLoP). 2009.

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New Arts, Commerce and Science College, Ahmednagar
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Syllabus
M.Sc. Computer Science-II

Title of the Course: Lab Course on Full Stack Development								
Year: I				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-15	MS-CS633P	00	03	03	60	30	70	100

Learning Objectives:

- Learn to research new methods of development in web applications and programming languages.
- Prepare mock-ups and storyboards for a web development project.
- Consult with clients to develop and document web site requirements.
- Demonstrate communication skills, service management skills, and presentation skills.

Course Outcomes (Cos)

- Use their learned skills, knowledge and abilities to develop web sites for the internet.
- Apply basic design principles to present ideas, information, products, and services on websites.

Detailed Syllabus:

Below is an example outline for a lab book covering some of the mentioned topics. Each topic has a brief description, practical exercises, and suggested assignments.

Lab Book Outline:

1. HTML5 and CSS3:

Topic Overview: Introduction to HTML5 and CSS3 for structuring and styling web pages.

Practical Exercises:

- Create a basic HTML5 document with appropriate semantic elements.
- Style the document using CSS, applying different layout techniques.
- Build a responsive webpage using media queries.

Assignments:

1. Design and code a personal portfolio webpage with multiple sections.
2. Create a webpage that demonstrates a responsive grid layout for imagegalleries.
3. Build a form with HTML5 validation and style it using CSS.

2. JavaScript:

Topic Overview: Fundamentals of JavaScript for dynamic and interactive web development.

Practical Exercises:

- Write JavaScript functions for basic arithmetic operations.

- Manipulate the DOM using JavaScript to create dynamic content.
- Implement an asynchronous task using Promises and Async/Await.

Assignments:

1. Develop a simple interactive quiz application using JavaScript.
2. Create a dynamic image gallery with JavaScript event handling.
3. Build a weather application that fetches data asynchronously from an API.

3. Front-end Frameworks (React.js):

Topic Overview: Introduction to React.js for building scalable and modular user interfaces.

Practical Exercises:

- Create a React component and render it on a webpage.
- Manage state and props in a React application.
- Implement a simple React router for navigation.

Assignments:

1. Build a ToDo list application using React with CRUD operations.
2. Develop a dynamic user interface that fetches data from a RESTful API.
3. Create a React application with state management using Redux.

4. Node.js and Express.js:

Topic Overview: Server-side development with Node.js and Express.js.

Practical Exercises:

- Set up a basic Node.js server.
- Create routes and handle HTTP requests using Express.js.
- Integrate a simple database (e.g., MongoDB)

5. Node.js.Assignments:

1. Build a RESTful API with CRUD operations using Node.js and Express.js.
2. Develop a user authentication system with JWT tokens.
3. Create a server-side application that interacts with a database and servesdata to a React front end.

6. Full-Stack Integration and Development:

Topic Overview: Integration of front-end and back-end components and deployment strategies.

Practical Exercises:

- Connect a React front end to an Express back end.
- Implement containerization using Docker.
- Set up a basic CI/CD pipeline for automated deployment.

Assignments:

1. Build and deploy a full-stack application on a cloud platform (e.g., AWS,Heroku).
2. Containerize an application and deploy it using Kubernetes.
3. Implement continuous integration with automated testing for a full-stackapplication.

Note:

Assignments should be designed to reinforce the concepts learned in each topic and progressively challenge students. Additionally, students can be encouraged to customize their projects based on their interests and career goals.

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Syllabus

M.Sc. Computer Science-II

Title of the Course: Lab Course on Machine Learning								
Year: I				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-16	MS-CS634P	00	03	03	60	30	70	100

Learning Objectives:

- Understand the mathematical and statistical prospective of machine learning algorithms through python programming.
- Design and evaluate the unsupervised models through python in built functions.
- Evaluate the machine learning models pre-processed through various feature engineering algorithms by python programming.

Course Outcomes:

- Able to use specific frameworks as per applications need.
- Design java application using design pattern techniques.
- Process available data using python libraries and predict outcomes using Machine Learning algorithms to solve given problem.
- Able to estimate Machine Learning models efficiencyusing suitable metrics

Sr. No	Assignments
1.	Write a python program to Prepare Scatter Plot (Use Forge Dataset /Iris Dataset)
2.	Write a python program to find all null values in a given data set and remove them
3.	Write a python program the Categorical values in numeric format for a given dataset
4.	Write a python program to implement simple Linear Regression for predicting house price
5.	Write a python program to implement multiple Linear Regression for a given dataset
6.	Write a python program to implement Polynomial Regression for given dataset
7.	Write a python program to Implement Naïve Bayes
8.	Write a python program to Implement Decision Tree whether or not to play tennis

9. Write a python program to implement linear SVM
10. Write a python program to find Decision boundary by using a neural network with 10 hidden units on two moons dataset
11. Write a python program to transform data with Principal Component Analysis (PCA)
12. Write a python program to implement k-nearest Neighbors ML algorithm to build prediction model (Use Forge Dataset)
13. Write a python program to implement k-means algorithm on a synthetic dataset
14. Write a python program to implement Agglomerative clustering on a synthetic dataset.
15. Data Sets for ML - UCI Machine Learning Repository - www.kaggle.com

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Syllabus

M.Sc. Computer Science-II

Title of the Course: Software Project Management								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-5	MS-CS635T	02	00	02	30	15	35	50

Learning Objectives:

- Identify the different project contexts and suggest an appropriate management strategy.
- Practice the role of professional ethics insuccessful software development.
- Identify and describe the key phases of project management.
- Determine an appropriate project management approach through an evaluation of the business context and scope of the project.

Course Outcomes:

- Identify the different project contexts and suggest an appropriate management Strategy
- Identify and describe the key phases of project management.
- Determine an appropriate project management approach through anevaluation of thebusiness context and scope of the project.

Sr.No	**Course Contents**	**Allotted Hours**
Unit I	**Introduction to Project Management** 1.1 What is a Project? 1.1.1 What is Project management? 1.1.2 Project phases and project life cycle 1.1.3 Organizational structure 1.2 Qualities of Project Manager 1.3 Project Components 1.3.1 Project Integration Management-Project plan development and execution 1.3.2 Change controls 1.3.3 Configuration management	(03)

Unit II	**Project Scope, Time and Cost Management** 2.1 Scope Management 2.1.1 Strategic planning 2.1.2 Scope planning, definition 2.1.3 Verification and control 2.2 Time management 2.2.1 Activity planning 2.2.2 Schedule development and control 2.3 Cost Management 2.3.1 Cost estimation and Control	(04)
Unit III	**Software Quality Assurance** 3.1 Quality planning 3.2 Quality Assurance	(04)
Unit IV	**Human Resource and Communication Management** 4.1 Organizational planning 4.2 Staff acquisition 4.3 Information distribution 4.4 Reporting	(04)
Unit V	**Risk Management** 5.1 Risk and Risk Types 5.2 Risk Management Process 5.2.1 Risk Analysis 5.2.2 Risk Planning 5.2.3 Risk Monitoring 5.3 Quantification and control	(04)
Unit VI	**Procurement Management** 6.1 Solicitation 6.2 Contract administration	(04)
Unit VII	**Software Metrics** 7.1 The scope of software metrics 7.2 Software metrics data collection 7.3 Analyzing software data 7.4 Measuring size, structure, external attributes	(04)
Unit VIII	**Planning a measurement program** 8.1 What is metrics plan? 8.2 Developing goals, questions and metrics 8.3 Where and When: Mapping measures to activities 8.4 How: Measurement tools 8.5 Who: Measurers , analyst, tools revision plans	(05)
Unit IX	**Software Reliability and Quality Standards** 9.1 Software Reliability 9.1.1 Measurement and prediction 9.1.2 Resource measurement 9.1.3 Productivity, teams and tools 9.2 Quality Standards 9.2.1 CMM 9.2.2 PSP/TSP	(04)

Suggested Readings

1. Information Technology Project Management, 6th Edition Kathy Schwalbe ISBN-13

:9781111221751 , Cenage Learning

2. Software Metrics: A rigorous and Practical Approach by Norman E. Fenton and ShariLawrencePfleeger, International Thomson Computer Press
3. Software Engineering: A Practioner's Approach by Roger S. Pressman ISBN:9780071267823
4. Practical Software Metrics for Project Management and Process Improvement Robert B.Grady,Prentice hall, ISBN : 9780137203840

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New Arts, Commerce and Science College, Ahmednagar

(Autonomous)

Syllabus

M.Sc. Computer Science-II

Title of the Course: Case Studies Using Software Project Management								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-6	MS-CS636P	00	02	02	60	15	35	50

Learning Objectives:

- Identify the different project contexts and suggest an appropriate management strategy.
- Practice the role of professional ethics in successful software development.
- Identify and describe the key phases of project management.
- Determine an appropriate project management approach through an evaluation of the business context and scope of the project.

Course Outcomes:

- Identify the different project contexts and suggest an appropriate management Strategy
- Identify and describe the key phases of project management.
- Determine an appropriate project management approach through an evaluation of the business context and scope of the project.

Detailed Syllabus:

Software Project Management Lab Course Manual:

	Documentation relative for case study	Date and Sign of Teacher
Assignment 1: **Case study:** **Subject:**	**1. Introduction** 1.1 Background 1.2Objectives	
Assignment 2: **Case study:** **Subject:**	1.3 Purpose and scope 1.3.1 Purpose 1.3.2 Scope	
Assignment 3: **Case study:** **Subject:**	**2. System Analysis** 2.1 Existing System 2.2 Proposed System	
Assignment 4: **Case study:** **Subject:**	2.3 Requirement Analysis 2.4 Hardware Requirement 2.5 Software Requirement	
Assignment 5: **Case study:** **Subject:**	2.6 Justification of selection of technology **3. System Design** 3.1 Module division 3.2 Data Dictionary 3.3 ER Diagrams 3.4 DFD/UML Diagrams **4. Results and Discussion** **5. Conclusion and Future work** **6. References**	

Note:

- **Subject should given by the respective teacher to students.**
- **Prepare at least group of five students for 1 case study.**

Suggested Readings

1. Information Technology Project Management, 6th Edition Kathy Schwalbe ISBN-13 :9781111221751 , Cenage Learning
2. Software Metrics: A rigorous and Practical Approach by Norman E. Fenton and ShariLawrencePfleeger, International Thomson Computer Press
3. Software Engineering: A Practioner's Approach by Roger S. Pressman ISBN:9780071267823
4. Practical Software Metrics for Project Management and Process Improvement Robert B.Grady,Prentice hall, ISBN : 9780137203840

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New Arts, Commerce and Science College, Ahmednagar

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Syllabus

M.Sc. Computer Science-II

Title of the Course: Project								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
PR-1	MS-CS637P	00	04	04	Term Work	30	70	100

Detailed Syllabus:

The Project can be platform, Language and technology independent. Project will be evaluated by project guide. Assessment will be done weekly in the respective batch. Evaluation will be on the basis of weekly progress of project work, progress report, oral, results and documentation and demonstration. You should fill your status of the project work on the progress report and get the Signature of project guide regularly. Progress report should sharply focus how much time you have spent on specific task. report should sharply focus how much time you have spent on specific task.

SN	From Date To Date	Details of Project work

Head,

Dept. of Computer Science

Project Guideline:

Students should prepare design document using SE/UML techniques depends on yourproject • Project Report Contents should be as follows:

1. College certificate
2. Acknowledgement
3. Problem Definition
4. Existing System and need for the new system
5. Scope of the work
6. Feasibility study (Including H/W & S/W setup requirements)
7. Requirement Analysis (including fact finding methods used)

8. E-R diagrams
9. Decision trees/Decision tables
10. Normalized Database Design & Data Dictionary.
11. Data flow Diagrams (if applicable)
12. Use-case Diagrams
13. Class Diagrams
14. Object Diagrams
15. Sequence Diagrams
16. Collaboration Diagram
17. Activity Diagram
18. State Chart (if applicable)
19. Component Diagram
20. Deployment Diagram (if applicable)
21. User interface design Menus Input Screens using sample data Reports, Graphs using sample data
22. Testing & Implementation plan (Should contain testing strategies, techniques used & implementation approach used.)
23. User manual
24. Drawbacks, Limitations & Proposed enhancement
25. Abbreviations used (if any)
26. Bibliography/Reference (Including book titles, authors name, editions, publications, etc.)

Project Related Assignments:

Assignment 1

Assignment 2

Assignment 3

Assignment 4

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Syllabus

M.Sc. Computer Science-II

Title of the Course: Course Work								
Year: II				**Semester: IV**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-17	MS-CS641P	00	06	06	90	45	105	150

Course Work

Reading Course with Guide For active participation and academic development final year students must prepared himself/herself by reading reference books/suggested readings by respective guide.

Guide will assign topic/ subject to the learner. The topic suggested is related with technology and related with full time industrial project work or learner contribution in collecting reference material, understanding the topic of the reading course and accordingly prepare the topic or the subject through self learning mode.

30% weightage should be given to punctuality, enthusiasm and aptitude of the students. Remaining, 70 % weightage for written examination may for this subject be conducted by respective guide

During IT period & completing reading course, it is desirable that student may present paper /poster in any conference. That will help to prove enthusiasm.

To earn the credit in this course is compulsory to complete the degree.

IA (45 marks)				
Weekly Attendance	Weekly Reports	First Presentation	Second Presentation	Documentation
05	05	10	10	15

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New Arts, Commerce and Science College, Ahmednagar

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Syllabus

M.Sc. Computer Science-II

Title of the Course: Full time Industrial Training								
Year: II				**Semester: IV**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
PR-02	MS-CS644P	00	16	16	Term Work	100	300	400

Teaching Scheme for Industrial Training

2 hours/week

The Industrial Training /Institutional project is equivalent to 5 theory courses of 4 credits each. Marks per 4 credits = 100 .
02 credits for internal evaluation.
The total weightage for Industrial/Institutional training is 550 marks.
Each student must individually complete minimum 5 months full time Industrial training /Institutional project in the 4th semester

Workload:

One mentor to be assigned for 6 students.

1 hours /week to be allotted for 6 students

Guidelines:

- Each student must individually complete **minimum 5 months** full time Industrial training /Institutional project in the 4th semester.
- College should assign a student mentor to every student. The mentor will monitor the progress of the student throughout the semester for continuous assessment.
- Student should submit a valid offer letter and synopsis within two weeks of starting the internship.
- There will be continuous assessment of the work done by the student during the internship period.

Continuous assessment guidelines:

1. Student should submit a weekly report in the college to the mentor.
2. The report should contain the following details: Name of student, project title, company name, company mentor, daily activities and results/output, proposed work for next week.
3. The weeklyreport should be duly signed by the student and company mentor/ institute guide(CM).
4. Student Mentor should maintain weekly attendance record for every student.
5. Two presentations should be conducted for each student (first presentation after

first month and second presentation after 3rd month)

6. Student Mentor should take feedback from the Company mentor regarding overall performance of the student.
7. At the end of the internship period, each student should prepare a report which should conform to international academic standards.
8. The report should follow the style in academic journals and books, with contents such as: abstract, background, aim, design and implementation, testing, conclusion and full references.

   Tables and figures should be numbered and referenced to in the report.

Examination and Evaluation guidelines

The project done during internship period will be evaluated in the following manner: IA - 150 marks + UE-400 marks.

The final presentation and documentation will be evaluated by three examiners:

1. Student mentor (appointed by respective college)
2. External examiner (appointed by the University)
3. IT expert (appointed by respective college)

IA (100 marks)				
Weekly Attendance	Weekly Reports	First Presentation	Second Presentation	Documentation
10	30	15	15	30

UE (300 marks)		
Mentor	IT Expert	External Examiner
100	100	100

Recommended Documentation contents:

Title page
Company / Institute
certificate
Internship
completion
certificate

Abstract
Introduction

-motivation
-problem statement
-purpose/objective and goals
-literature survey
-project scope and limitations

System analysis

-Comparative study of Existing systems
- scope and limitations of existing systems
-project perspective, features
- stakeholders
-Requirement analysis - Functional requirements, performance requirements, security requirements etc.

System Design

- Design constraints
- System Model: UML diagrams
- Data Model
- User interfaces

Implementation details

- Software/hardware specifications, etc.

Reports Testing

Test Plan, Black Box Testing or Data Validation Test Cases, White Box Testing or Functional Validation Test cases and results

Conclusion and Recommendations & Future Scope

Bibliography and Reference